

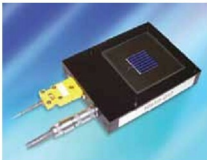
nation velocity (S) and minority carrier diffusion length (L_n) are measured under this category.

Dark I-V solar cell testing

Highly accurate characterisation of solar cell's resistance and diode properties are measured by this test. By blocking all light, the PV cell can be tested as a passive diode element to determine its breakdown diode properties and internal resistances. This type of testing is performed to ensure that the quality of the cell meets the application's criteria and the cell is defect-free.

Such thorough testing of the solar cell is often done in space applications, where verifying quality and reliability is essential. Testing a solar cell in a dark chamber causes the cell to behave essentially as a diode with some resistance characteristics. With no light stimulus applied to the solar cell, dark testing requires a DC source that only has the ability to source or output current. Dark testing does require both positive and negative voltage to test the I-V curve. The negative voltage is used to reverse the cell's bias, which approximates the parallel resistance of the cell (R_p) and the breakdown region of the diode.

The measurement set up for measuring the I-V curve perfor-



The Oriel reference cell

mance of cells, modules and arrays is built with power supplies, switch/measurement units, electronic loads and digital multimeters. The set up should be capable of measuring the I-V curve performance of cells, modules and arrays in both forward and reverse polarities, in full radiation, and in shadow (darkness).

DC power source. A DC power supply capable of providing voltage in forward and reverse polarities is needed for measurement of the performance of a solar cell or module. The power supply acts as a variable voltage load when the solar device is illuminated, which enables measurement of the power I-V curve. When the solar device is dark, voltage is applied in both forward and reverse directions and you can measure the entire dark I-V curve.

Complete characterisation of a solar device generally requires a four-quadrant-capable power source. A four-quadrant DC source, capable of negative voltages and negative currents is often used for solar cell testing for two reasons. The first is to overcome any series resistance in the cell; the second reason is that using negative voltage values may be desirable to reverse the cell's bias to fully characterise the cell's electrical properties.

Measuring the I-V performance of an illuminated solar device requires the power supply to sink current while sourcing positive voltage. Measuring the I-V performance of a dark cell requires the power supply to source positive and negative voltage and

current to cover the full range of solar device operation.

A bipolar power supply is generally the instrument of choice. If a bipolar source is not available, a pair of two-quadrant power supplies configured appropriately can be a viable alternative. Power supplies 661xC and 663xB of Agilent Technologies among other OEMs are typical examples of DC power source.

Electronic loads. Electronic loads are solutions for solar module testing because of their wide power range and ability to sink large amounts of current. Electronic loads typically have three modes of operation—constant current (CC), constant voltage (CV) and constant resistance (CR). CV is the preferred mode of operation for I-V curve tracing because it allows stepping through voltages incrementally and measure the current output of the module under test.

For testing outdoors, or in other places where environmental temperatures vary widely, an electronic load that provides temperature coefficient specifications is highly desirable. Variations in environmental temperatures change the measurement specifications of the load's built-in measurements, leading to increased measurement uncertainty. Temperature coefficient specifications allow for compensation of environments with wide temperature variations.

Switch and measurement instrument. In solar cell and module testing, measurement of parameters other than I-V curve such as temperature and calibrated reference cells are also required. Temperature has a direct effect on the output power of the cell or module. Calibrated reference cells are often needed to gauge the effectiveness of the light source used to power the solar cell or module. A switching configuration can also be used to allow multiple solar cells or modules to be tested in parallel.

Reference cells. These are an integral part of solar simulator calibration and solar cell I-V characterisation.



I-V 400 multimeter